

Octubre 1 2020

Juan José Pestrepa Rosero

Santiago Hurtado Bustos

Taller

① Encontrar los puntos sobre la superficie $XYZ = 8$

$$XYZ = 8$$

origen $(0,0,0)$

Punto $P(x,y,z)$

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + \dots}$$

$$d(0,P) = \sqrt{x^2 + y^2 + z^2}$$

$$d^2 = x^2 + y^2 + z^2$$

Función a minimizar: $f(x,y,z) = x^2 + y^2 + z^2$

Condición: $XYZ = 8$

$$g(x,y,z) = 0$$

$$XYZ = 8$$

$$XYZ - 8 = 0$$

$$\nabla f(x,y,z) = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\nabla g(x,y,z) = yz\hat{i} + xz\hat{j} + xy\hat{k}$$

$$\nabla f(x,y,z) = \lambda \nabla g(x,y,z)$$

$$2x\hat{i} + 2y\hat{j} + 2z\hat{k} = \lambda yz\hat{i} + \lambda xz\hat{j} + \lambda xy\hat{k}$$

$$2x = \lambda y z$$

$$2y = \lambda x z$$

$$2z = \lambda x y$$

$$\lambda = \frac{2x}{yz}$$

$$2y = \frac{2x}{yz} \times z$$

$$\sqrt{y^2} = \sqrt{x^2}$$

$$y = \pm x$$

$$2z = \frac{2x}{yz} \times y$$

$$\sqrt{z^2} = \sqrt{x^2}$$

$$z = \pm x$$

$$\begin{array}{c|c} x & x \\ x & x \\ x & x \\ x & x \\ x & x \end{array}$$

$$xyz = 8$$

$$\bullet y = x, z = x$$

$$x \cdot x \cdot x = 8$$

$$\sqrt[3]{x^3} = \sqrt[3]{8}$$

$$\boxed{x = 2}$$

$$P(2, 2, 2)$$

$$\bullet y = -x, z = x$$

$$x \cdot (-x) \cdot x = 8$$

$$-x^3 = 8$$

$$\sqrt[3]{x^3} = \sqrt[3]{-8}$$

$$x = -2$$

$$P(-2, 2, -2)$$

$$\bullet y = x, z = -x$$

$$x \cdot x \cdot (-x) = 8$$

$$-x^3 = 8$$

$$x = -2$$

$$P(2, 2, -2)$$

$$\bullet y = -x, z = -x$$

$$x \cdot (-x) \cdot (-x) = 8$$

$$\sqrt[3]{x^3} = \sqrt[3]{8}$$

$$x = 2$$

$$P(2, -2, -2)$$

② Evalúe la integral

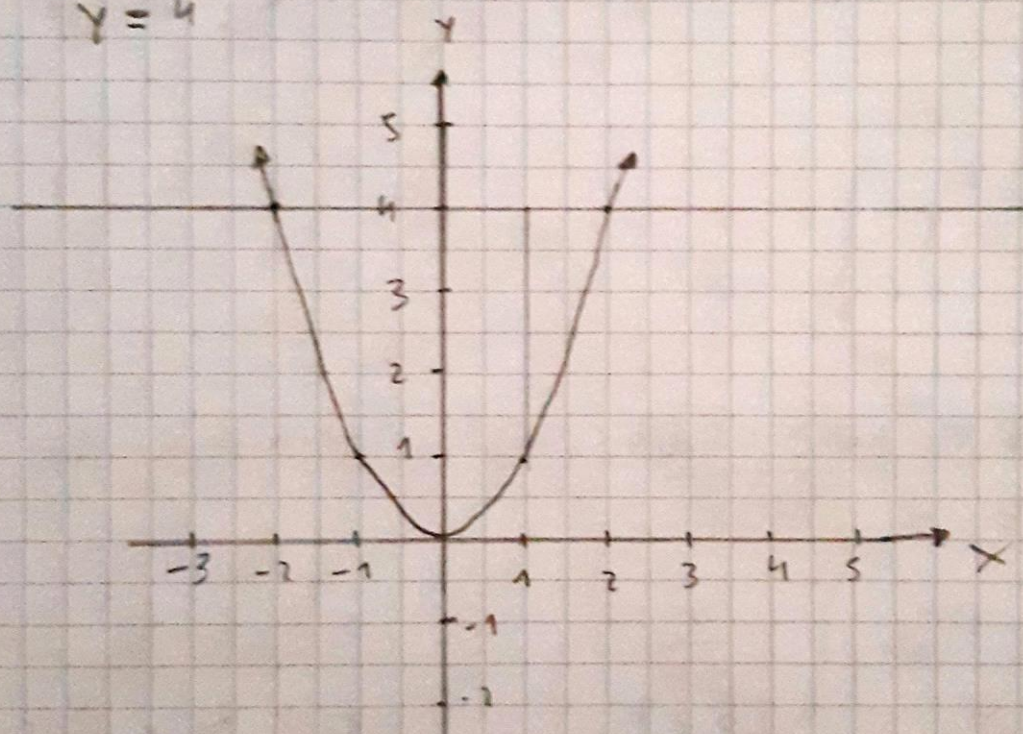
$$\iint_R 4xe^{y^2} dA$$

Integ
Log
Exp
Trig
K=8

Región comprendida entre:

$$y = x^2 \rightarrow x = \sqrt{y}$$

$$y = 4$$



$$0 \leq x \leq \sqrt{y}$$

$$0 \leq y \leq 4$$

$$\int_0^4 \int_0^{\sqrt{y}} 4xe^{y^2} dx dy$$

$$4 \int_0^4 \frac{y}{2} e^{y^2} dy$$
$$2 \int_0^4 ye^{y^2} dy$$

$$= 4 \int_0^4 \left(\frac{x^2}{2} e^{y^2} \right) \Big|_0^{\sqrt{y}} dy$$

$$u = y^2 \quad \sqrt{u} = y$$

$$du = 2y dy$$

$$\frac{du}{2y} = dy$$

$$2 \int_0^4 e^u \frac{y}{2y} du$$

$$= \frac{2}{2} \int_0^4 e^u dy$$

$$= \left(e^u \right)_0^4$$

$$= (e^{16} - 1)$$

b) calcular el volumen del sólido determinado por

$$f(x, y) = x \sqrt{y^3 + 1}$$

Plano x, y

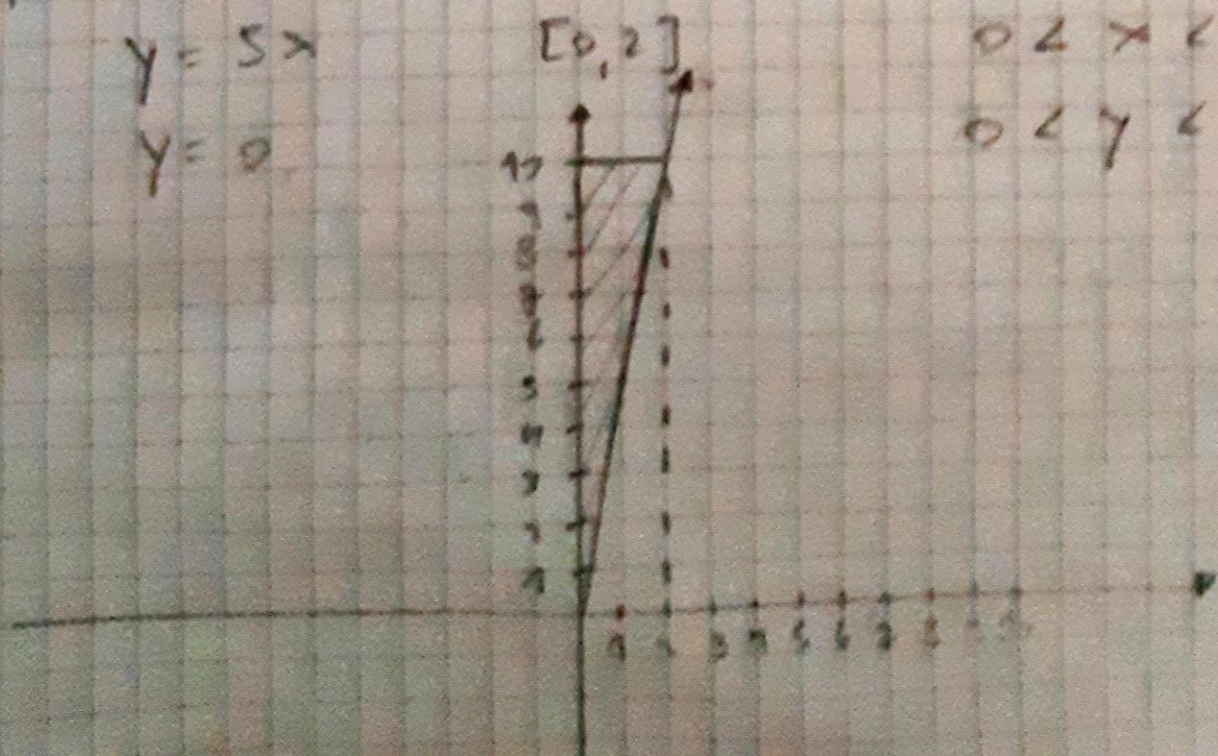
$$y = 5x$$

$$y = 0$$

$[0, 2]$

$$0 < x < 4/5$$

$$0 < y < 10$$



$$V = \int_0^{10} \int_0^{y/5} x \sqrt{y^3 + 1} \, dx \, dy$$

$$= \int_0^{10} \left[\frac{x^2 \sqrt{y^3 + 1}}{2} \right]_0^{y/5} dy$$

$$= \int_0^{10} \left(\frac{\frac{y^2}{25} \sqrt{y^3 + 1}}{2} \right) dy$$

$$= \int_0^{10} \frac{y^2}{50} \sqrt{y^3 + 1} \, dy = \frac{1}{50} \int_0^{10} y^2 \sqrt{y^3 + 1} \, dy$$

$$u = y^3 + 1$$

$$du = 3y^2 \, dy$$

$$dy = \frac{1}{3y^2} du$$

$$= \frac{1}{50} \int_0^{10} \frac{\sqrt{u}}{3} \, du$$

$$= \frac{1}{150} \int_0^{10} u^{1/2} \, du$$

$$= \frac{1}{150} \left(\frac{2u \sqrt{u}}{3} \right)$$

$$= \frac{1}{150} \left(\frac{(2y^3 + 2) \sqrt{y^3 + 1}}{3} \right) \Big|_0^{10}$$

$$= \frac{1}{150} \cdot \left(\frac{(2002) \cdot \sqrt{1001}}{3} - \frac{2}{3} \right) = \frac{(2002) \cdot \sqrt{1001} - 2}{450}$$

③ Eje $x \rightarrow$ Region

$$y = e^x \quad [0, 1]$$

$$\rho(x, y) = y^3$$

$$m = \int_0^1 \int_0^{e^x} y^3 dy dx$$

$$= \int_0^1 \left(\frac{y^4}{4} \Big|_0^{e^x} \right) dx$$

$$= \int_0^1 \left(\frac{e^{4x}}{4} \right) dx = \frac{1}{4} \left(\frac{e^{4x}}{4} \Big|_0^1 \right)$$

$$= \frac{1}{4} \left(\frac{e^4}{4} - \frac{1}{4} \right)$$

$$m = \frac{e^4 - 1}{16}$$

$$M_x = \int_0^1 \int_0^{e^x} y^4 dy dx$$

$$= \int_0^1 \frac{e^{5x}}{5} dx$$

$$= \int_0^1 \left(\frac{y^5}{5} \Big|_0^{e^x} \right) dx = \frac{1}{5} \left(\frac{e^{5x}}{5} \Big|_0^1 \right)$$

$$= (e^5 - 1) \frac{1}{25} = \frac{e^5 - 1}{25}$$

$$M_y = \int_0^1 \int_0^1 x y^2 dy dx$$

$$= \int_0^1 \left(x \frac{y^3}{3} \Big|_0^1 \right) dx$$

$$u = y$$

$$dv = e^{ux} dx$$

$$du = dy$$

$$v = \frac{e^{ux}}{u}$$

$$= \frac{1}{3} \int_0^1 x e^{ux} dx = x \frac{e^{ux}}{u} - \int \frac{e^{ux}}{u} dx$$

$$= \frac{1}{u} \left(\frac{x e^{ux}}{u} - \frac{1}{u} \int e^{ux} dx \right)$$

$$= \frac{1}{u} \left(\frac{x e^{ux}}{u} - \frac{1}{16} e^{ux} \Big|_0^1 \right)$$

$$= \frac{1}{u} \left(\frac{e^u}{u} - \frac{e^u}{16} \right) - \left(-\frac{1}{16} \right)$$

$$= \frac{1}{u} \left(\frac{16 e^u - u e^u}{64} + \frac{1}{16} \right)$$

$$= \frac{1}{u} \left(\frac{12 e^u}{64} + \frac{1}{16} \right)$$

$$= \frac{1}{u} \left(\frac{3 e^u}{16} + \frac{1}{16} \right) = \left(\frac{3 e^u + 1}{16} \right) \frac{1}{u}$$

$$= \frac{3 e^u + 1}{64}$$

$$\bar{y} = \frac{e^5 - 1}{25} \div \frac{e^4 - 1}{16} = \frac{16}{25} \frac{(e^5 - 1)}{(e^4 - 1)} = 1,76$$

$$\bar{x} = \frac{3e^4 + 1}{64} \div \left(\frac{e^4 - 1}{16} \right) = \frac{16}{64} \frac{(3e^4 + 1)}{(e^4 - 1)} = \frac{3e^4 + 1}{4e^4 - 4} = 0,768$$

Centro Masa $\left(\underset{\bar{x}}{0,768}, \underset{\bar{y}}{1,76} \right)$

(4) a) $\int_0^1 \int_0^{\sqrt{1-y^2}} e^{x^2+y^2} dx dy$

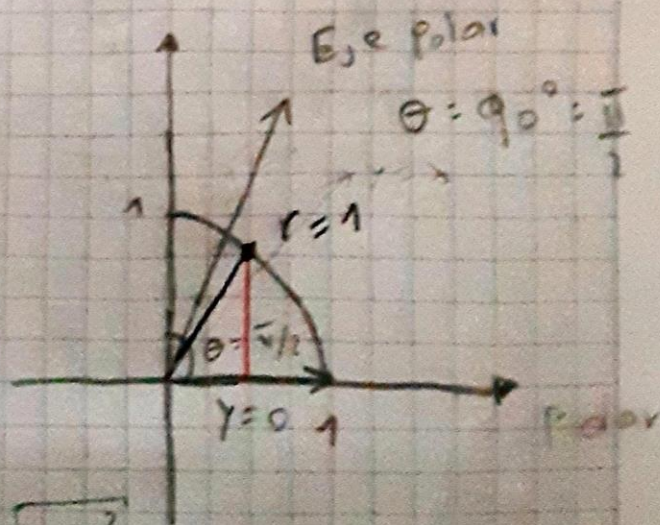
$$r^2 = x^2 + y^2, \quad r < 1$$

$$y = \sqrt{r^2 - x^2}$$

$$y = \sqrt{1 - x^2}$$

$$0 < y < \sqrt{1-x^2}$$

$$0 < x < 1$$



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\int_0^{\pi/2} \int_0^1 e^{r^2} r dr d\theta = \int_0^{\pi/2} \left(\frac{1}{2} e^{r^2} \right)_0^1 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} (e - 1) d\theta = \frac{1}{2} \left(e\theta - \theta \right)_0^{\pi/2}$$

$$= \frac{1}{2} e \frac{\pi}{2} - \frac{\pi}{2} = \left(\frac{e\pi - \pi}{2} \right) \frac{1}{2}$$

$$= \frac{1}{4} \pi (e - 1)$$

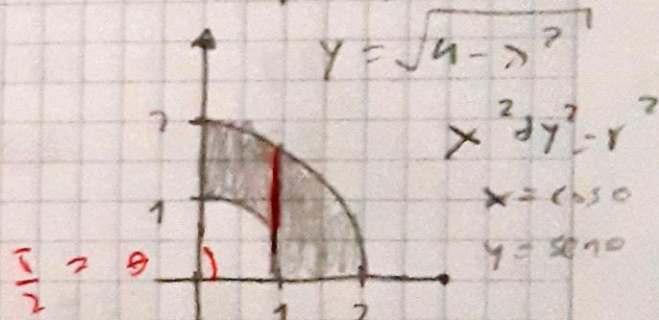
$$x^2 + y^2 = 4$$

$$x^2 + y^2 = 1$$

$$y = \sqrt{1 - x^2}$$

$$y = \sqrt{4 - x^2}$$

$$b) \int_0^1 \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} \frac{x^2}{x^2 + y^2} dy dx$$



$$= \int_0^{\pi/2} \int_1^2 \frac{r^2 \cos^2 \theta}{r^2} r dr d\theta$$

$$= \int_0^{\pi/2} \int_1^2 r \cos^2 \theta dr d\theta$$

$$= \int_0^{\pi/2} \left(\frac{r^2}{2} \right)_1^2 \cos^2 \theta d\theta$$

$$= \int_0^{\pi/2} \left(2 - \frac{1}{2} \right) \cos^2 \theta d\theta$$

$$= \int_0^{\pi/2} \frac{3}{2} \cos^2 \theta d\theta$$

$$= \frac{3}{2} \int_0^{\pi/2} \cos^2 \theta d\theta$$

$$\frac{3}{2} \left(\frac{1}{2} \theta + \frac{\sin(2\theta)}{4} \right)_0^{\pi/2} = \frac{3}{2} \left(\frac{\pi}{4} \right) = \frac{3\pi}{8}$$